CampusSense: Local Nano AI Engine & UI Blueprint

COESC x OCTA : Hack4Success 2026 | Track: Student Around Services (LSPU-SPCC)

> **Architecture Selected: Path 3 (Hybrid Edge-First Nano RAG)**
>
> **Core Philosophy:** On-device local intelligence handles student queries offline in under 15ms with zero cloud cost, optimized for budget devices (Android 12, Snapdragon 460, 3-4GB RAM). An optional cloud adapter provides multi-tier AI expansion.

1. System Architecture & Query Pipeline

The on-device Nano AI implements an offline **Retrieval-Augmented Generation (RAG)** pipeline with four lightweight stages:

Stage	Mechanism	Benefit for Hackathon
1. Intent & Tokenizer	Bilingual parser (English + Tagalog/Taglish) detects intents: `SUSPENSION`, `SCHOLARSHIP`, `ACADEMICS`, `EVENTS`.	Understands student slang (*"may pasok ba"*, *"walang klase"*, *"kailan pasahan"*).
2. Context Ranker	Hybrid scoring over local Room SQLite database: `Score = (IntentWeight × 0.4) + (KeywordHits × 0.4) + (Recency × 0.2)`	Prioritizes the freshest announcement from official pages (LSPU Official, SPCC OSAS).
3. Answer Synthesizer	Extracts operative facts (status, dates, guidelines) and compiles conversational responses.	Zero hallucinations; answers cite exact dates and page sources.
4. Source Citation	Attaches post permalink, page name, and timestamp.	Judges see verifiable proof of social media post authenticity.

2. Hardware & Performance Target

- **Target Hardware:** Android 12 (API 31), Qualcomm Snapdragon 460 (8× Kryo 240 cores), 3–4GB RAM.
- **Memory Consumption:** < 20 MB RAM during query execution.
- **Inference Latency:** < 15 milliseconds (sub-second instant response).
- **Network Dependency:** 0% for core queries (100% functional in Airplane Mode).

3. Class Interface Specifications

A. `LocalNanoAIEngine.java`

The main on-device engine class placed in `com.example.campussync.ai`:

```
package com.example.campussync.ai;

import com.example.campussync.data.local.AnnouncementEntity;
import java.util.List;

public class LocalNanoAIEngine {

    public enum Intent {
        SUSPENSION,     // Class suspensions, weather advisories
        SCHOLARSHIP,    // TDP, CHED, UniFAST, OSAS validation
        ACADEMICS,      // Exams, grading, enrollment, permits
        EVENTS,         // Campus activities, student council
        GENERAL,        // Greetings, help, bot capabilities
        UNKNOWN         // Fallback when no match is found
    }

    public static class NanoAIResult {
        public final String answer;
        public final Intent detectedIntent;
        public final float confidence;
        public final AnnouncementEntity primarySource;
        public final boolean isOfflineAnswer;

        public NanoAIResult(String answer, Intent intent, float confidence,
                            AnnouncementEntity source, boolean isOffline) {
            this.answer = answer;
            this.detectedIntent = intent;
            this.confidence = confidence;
            this.primarySource = source;
            this.isOfflineAnswer = isOffline;
        }
    }

    /**
     * Executes the on-device RAG resolution synchronously or via executor.
     */
    public NanoAIResult processQuery(String userQuery, List<AnnouncementEntity> cachedPosts)
        // 1. Normalize and extract intent
        // 2. Score candidate posts from Room DB
        // 3. Synthesize natural language answer with citation
        // Returns structured result ready for Chat UI
    }
}
```

B. Bilingual Intent Dictionaries (English + Taglish)

Intent	Keywords & Colloquial Trigger Patterns

SUSPENSION	pasok, klase, suspindido, walang, suspension, typhoon, bagyo, signal, cancel, cancelled
SCHOLARSHIP	scholarship, tdp, ched, osas, allowance, unifast, stipend, grant, validation
ACADEMICS	midterm, final, exam, permit, grades, enrollment, add/drop, schedule, registrar
EVENTS	event, intrams, octa, coesc, orientation, webinar, activity, general assembly

4. UI Navigation & Feed Updates (`MainActivity.java`)

Your UI responsibility includes rebranding and connecting the engine:

- **Branding:** Update all app references to **CampusSense** (*"Intelligent Campus Information Assistant"*).
- **Dynamic Header Badge:** Display `● NANO AI READY (OFFLINE)` in green (`#247A57`) to prove edge autonomy to judges.
- **Category Filter Pills on Feed:** Interactive filter pills:
  `[ All ]` `[ 🌧️ Suspensions ]` `[ 🎓 Scholarships ]` `[ 📚 Academics ]` `[ 📢 Events ]`
- **Chat Screen Hook:** When user taps Send or a Quick Prompt Chip, pass the text to `LocalNanoAIEngine.processQuery()` and render response with direct *"View Original Facebook Post ↗"* button.

5. Golden Test Cases for Live Demonstration

Student Query	Detected Intent	Expected Nano AI Output
"May pasok ba bukas sa LSPU?"	`SUSPENSION`	**"Classes remain normal as of today."** (or *"Classes are suspended tomorrow due to Tropical Cyclone Wind Signal #2 according to LSPU Official. Read post ↗"*)
"Kailan deadline ng TDP scholarship?"	`SCHOLARSHIP`	**"According to SPCC OSAS, the submission deadline for TDP documents is Friday, Nov 14 at 5:00 PM at the OSAS office. Read post ↗"**
"Midterm exam schedule"	`ACADEMICS`	**"Midterm examinations are scheduled on Nov 24–28 per the LSPU Academic Calendar advisory. Read post ↗"**
"Unknown question"	`UNKNOWN`	**"I couldn't find an official announcement matching that topic in your offline cache. Try searching official pages directly or sync when online."**

6. 10-Minute Live Pitch Climax Strategy (CTE Function Hall)

> **Stage Demo Sequence: The "Airplane Mode" Moment**
>
> 1. Mirror your phone to the screen using `scrcpy`.
> 2. Swipe down the Android notification shade on stage and turn on **AIRPLANE MODE**.
> 3. Type: *"May pasok ba bukas?"*
> 4. Show the immediate response generated in under 15ms with full source citation.
> 5. Pitch line: *"While traditional chatbots fail when campus Wi-Fi drops, CampusSense's Local Nano AI ensures no LSPU student ever misses a life-critical announcement."*